

KOZASAJ, Kazakhstan

WMO: 355290

Lat: 48.217N

Lon: 57.117E

Elev: 151

StdP: 99.52

Time Zone: 5.00 (E05)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.7	-26.1	-32.6	0.2	-28.6	-29.5	0.2	-25.9	16.1	-9.9	14.0	-9.6	0.7	50	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	35.6	21.0	33.7	20.1	31.9	19.5	23.0	32.9	21.7	30.9	20.4	29.4	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.6	27.7	18.5	13.6	26.5	17.0	12.4	25.4	69.0	32.4	63.9	31.1	59.4	29.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.1	9.0	-32.0	39.1	2.2	1.2	-33.5	39.9	-34.8	40.6	-36.0	41.3	-37.6	42.2		
(5)			-32.1	24.4	2.1	1.7	-33.7	25.7	-34.9	26.7	-36.1	27.7	-37.6	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-10.8	-13.3	-5.9	8.4	15.7	22.5	23.8	21.7	15.3	6.6	-3.1	-8.6	(6)
(7)		DBStd	14.32	7.25	7.27	6.40	5.35	4.32	4.07	3.38	3.71	4.65	4.64	6.40	6.36	(7)
(8)		HDD10.0	2994	644	652	493	93	7	0	0	0	11	126	392	578	(8)
(9)		HDD18.3	4915	902	885	751	301	103	8	2	12	109	365	642	836	(9)
(10)		CDD10.0	1582	0	0	0	43	184	374	427	364	171	19	0	0	(10)
(11)		CDD18.3	463	0	0	0	1	22	132	171	118	19	0	0	0	(11)
(12)		CDH23.3	5752	0	0	0	24	297	1734	2059	1348	284	5	0	0	(12)
(13)		CDH26.7	2520	0	0	0	3	67	846	974	557	73	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.4	3.7	3.7	3.7	3.5	2.9	2.7	2.8	2.8	2.7	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	368	34	30	39	34	38	29	27	14	16	25	37	44	(15)
(16)		PrecMax	592	67	71	113	95	98	122	94	43	59	52	81	104	(16)
(17)		PrecMin	202	3	1	0	7	3	1	1	1	2	0	1	10	(17)
(18)		PrecStd	101	18	17	25	21	25	25	20	10	12	13	19	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.7	13.1	26.2	30.3	38.5	37.7	35.5	31.1	24.1	9.6	3.5	(19)
(20)			MCWB	0.7	1.1	7.8	14.7	17.8	23.6	22.7	19.1	19.9	14.4	5.5	2.7	(20)
(21)		2%	DB	0.8	0.8	7.9	22.9	28.7	36.1	35.6	33.5	28.8	20.9	7.1	1.9	(21)
(22)			MCWB	-0.1	0.0	4.7	12.5	16.7	21.0	21.8	19.2	17.7	11.3	4.2	1.4	(22)
(23)		5%	DB	-0.1	-0.5	4.9	20.1	26.7	33.7	33.8	31.7	26.5	17.3	5.7	0.8	(23)
(24)			MCWB	-0.9	-1.2	2.8	11.1	15.2	19.7	20.8	18.9	15.9	9.6	3.0	0.0	(24)
(25)		10%	DB	-1.3	-2.8	2.7	17.7	24.7	31.6	31.8	30.0	24.5	14.6	4.1	-0.1	(25)
(26)			MCWB	-2.0	-3.6	1.0	10.1	14.5	19.1	20.2	18.4	15.0	8.3	1.8	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.3	7.9	15.5	19.5	24.5	25.3	23.1	20.6	14.6	6.8	2.8	(27)
(28)			MCD B	1.5	2.2	12.2	23.6	28.7	37.0	34.4	30.9	30.6	22.0	7.9	3.1	(28)
(29)		2%	WB	0.0	0.2	4.9	13.8	17.4	22.4	23.5	21.4	18.4	12.6	5.0	1.5	(29)
(30)			MCD B	0.7	0.8	7.3	21.6	26.6	33.8	33.5	29.5	27.0	18.5	6.6	1.9	(30)
(31)		5%	WB	-0.8	-1.1	2.8	12.2	16.4	21.1	22.2	20.0	16.9	10.7	3.4	0.3	(31)
(32)			MCD B	-0.2	-0.4	4.7	19.0	24.6	31.2	31.1	28.8	24.1	16.0	5.1	0.8	(32)
(33)		10%	WB	-1.9	-3.7	1.1	10.7	15.2	19.8	20.8	19.0	15.6	8.8	2.0	-0.9	(33)
(34)			MCD B	-1.3	-2.8	2.6	16.9	22.8	28.6	29.6	27.9	23.2	13.6	3.6	-0.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.6	9.0	11.1	12.3	13.6	13.1	13.1	13.3	12.1	7.3	7.3	(35)
(36)			MCDBR	6.3	6.0	9.1	14.8	15.0	16.1	15.4	15.2	16.7	16.9	9.7	6.1	(36)
(37)			MCWBR	6.2	5.8	6.1	7.6	7.2	7.6	7.6	7.0	9.3	9.9	7.3	5.8	(37)
(38)		5% WB	MCD BR	6.1	6.0	8.4	13.5	13.6	14.9	14.1	13.7	15.2	15.0	8.0	5.8	(38)
(39)			MCWBR	6.1	5.9	5.8	7.4	7.5	8.1	7.9	7.5	9.4	9.7	6.5	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.336	0.402	0.445	0.435	0.429	0.421	0.425	0.373	0.366	0.343	0.317	(40)	
(41)		taud	2.089	2.015	2.003	2.021	2.130	2.206	2.247	2.210	2.307	2.240	2.207	2.119	(41)	
(42)		Ebn at Noon	719	788	793	802	829	836	837	814	825	756	680	654	(42)	
(43)		Edh at Noon	89	121	146	159	148	138	131	130	107	97	80	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.59	3.64	4.85	6.20	6.88	6.65	5.85	4.44	2.60	1.29	0.91	(44)	
(45)		RadStd	0.25	0.27	0.39	0.50	0.50	0.47	0.38	0.30	0.27	0.29	0.15	0.16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	+0.65	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page